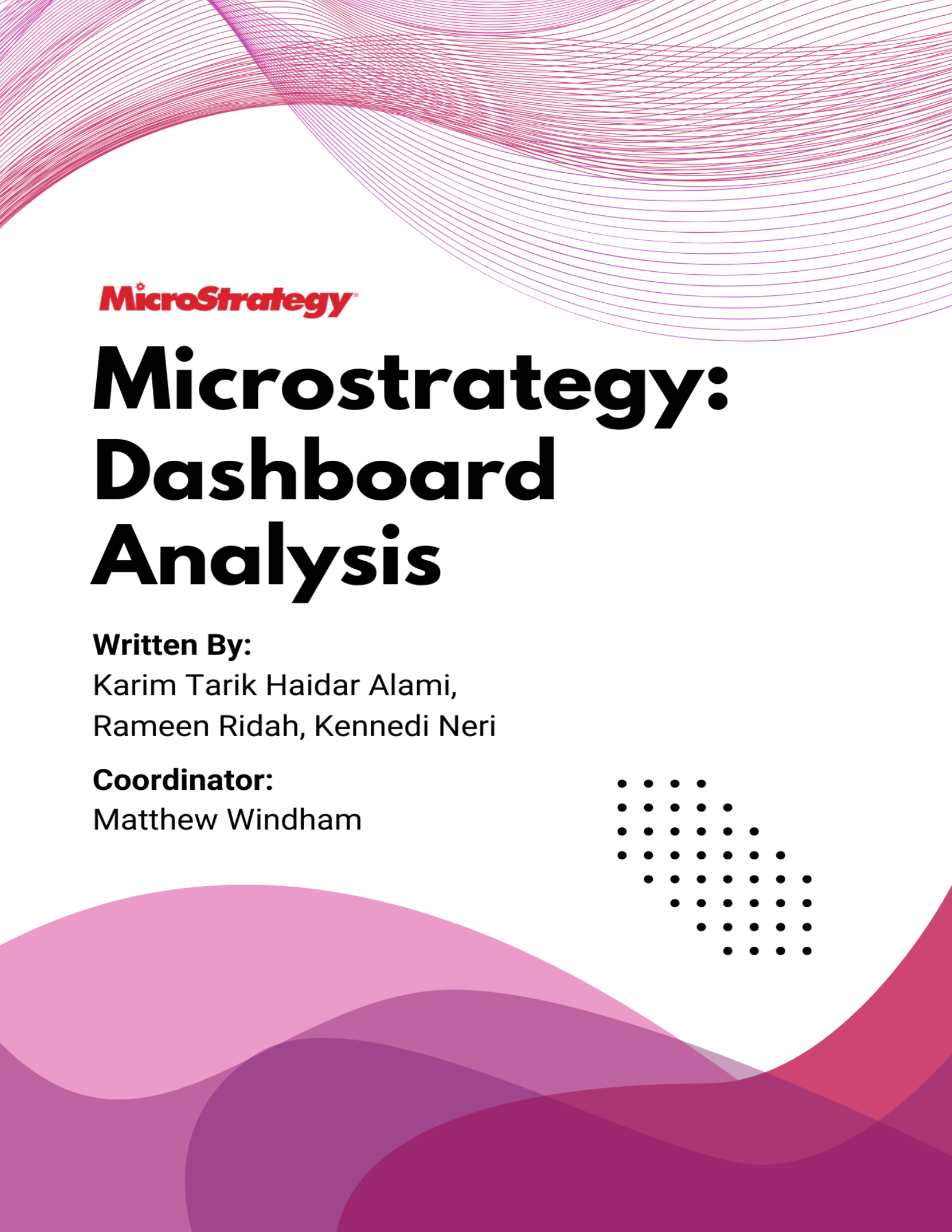




MicroStrategy

Microstrategy: Dashboard Analysis

Written By:

Karim Tarik Haidar Alami,
Rameen Ridah, Kennedy Neri

Coordinator:

Matthew Windham

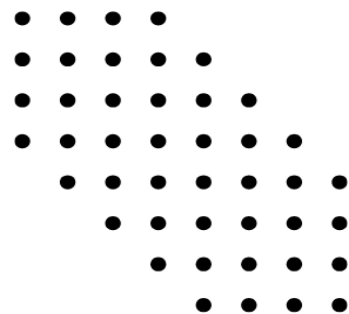


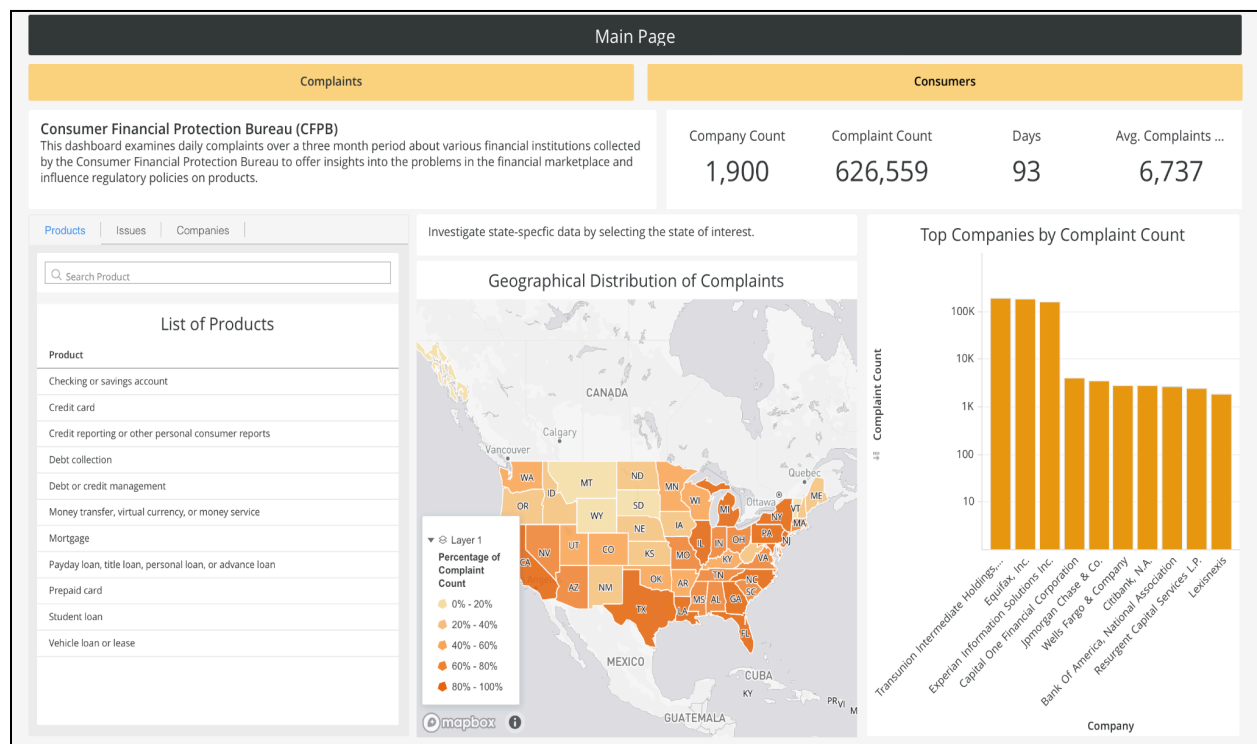
Table of Contents	2
Executive Summary	3
Why the CFPB Dataset?	5
The Problem We Set Out to Solve	5
Risks and Limitations	5
Building the Dashboard: Empowering Stakeholders	7
1. Main Page Overview	7
2. Complaint Analysis	9
3. Consumer Analysis	10
The Outcome: Actionable Insights	12
1. Highlighting High-Complaint Categories	12
2. Geographic Trends	12
3. Company Response Patterns	12
4. AI-Powered Insights	12
Why It Matters	13
1. For Consumers	13
2. For Policymakers	13
3. For Companies	13
What CFPB Can Do	13
1. Enhance Sentiment Tagging	13
2. Standardize Data Fields	13
3. Offer Consumer-Friendly Tools	13
Conclusion	14
Contact Information	14
Appendices	15

Executive Summary

MicroStrategy, a leader in business intelligence software, engaged our team to create an interactive dashboard for the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Dataset. This initiative aims to uncover trends and patterns in consumer complaints, providing actionable insights to improve transparency and decision-making.

Every year, thousands of consumers encounter challenges navigating the financial services industry—whether it is dealing with unresolved credit disputes, unexpected fees, or predatory lending practices. These concerns are voiced through complaints filed with the CFPB. While the dataset holds immense value for improving transparency and consumer protection, its sheer size and complexity makes it inaccessible to most.

Recognizing this opportunity, our team analyzed over **626,000 consumer complaints** to identify trends in financial product issues, particularly in **credit reporting** and **debt collection**. By combining Python for data analysis, MicroStrategy for visualization, and AI-powered insights, we transformed the raw dataset into an **actionable dashboard**.



Key Objectives

- **Develop an Interactive Dashboard:** Simplify access to insights for consumers, policymakers, and companies.
- **Analyze High-Complaint Categories:** Highlight issues like credit reporting and debt collection to inform action.
- **Provide Transparency and Accountability:** Equip stakeholders with tools to interpret the data effectively.

This project empowers stakeholders by making the CFPB data accessible and actionable. The dashboard supports **consumers** in understanding financial product risks, **policymakers** in addressing systemic issues, and **companies** in improving their customer service practices.

Why the CFPB Dataset?

The CFPB dataset is a direct reflection of consumer experiences—real voices highlighting real issues. However, the challenges are significant:

- The dataset contains hundreds of thousands of entries, making it overwhelming for manual analysis.
- Raw data lacks intuitive structure, making it difficult to identify trends or actionable insights.
- Without a user-friendly interface, its potential remains largely untapped for policymakers, companies, and consumers alike.

We set out to address these challenges with a clear mission: make the CFPB dataset accessible, understandable, and actionable for everyone.

The Problem We Set Out to Solve

While the CFPB publishes this data, stakeholders face major hurdles:

- **Consumers:** Lack the tools to compare companies, analyze trends, or identify high-risk financial products.
- **Policymakers:** Struggle to spot systemic issues like predatory lending or recurring complaints about specific practices.
- **Companies:** Miss opportunities to improve customer service by failing to analyze their own performance relative to competitors.

Our goal was to build a solution that simplifies data access, provides meaningful insights, and enables informed decision-making.

Risks and Limitations

The CFPB Dataset is a valuable resource for analyzing consumer grievances in the financial services industry. However, like any large dataset, it comes with several risks and limitations that may impact its use and the accuracy of insights derived from it. These considerations are critical for stakeholders interpreting the data and applying it to decision-making.

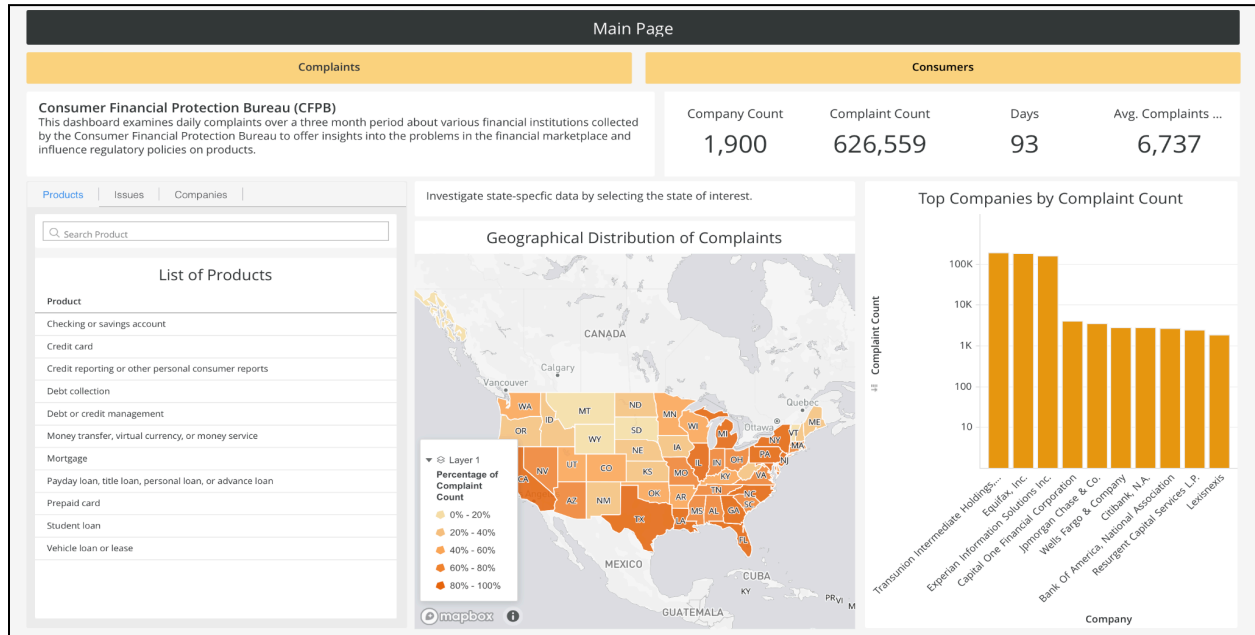
- **Uncaptured Informal Disputes:** Many grievances may be resolved informally between consumers and companies and are not included in the dataset, which could skew the perception of overall industry performance.
- **Regional Imbalances:** States with higher populations or more engaged consumers may report disproportionately more complaints, which might not accurately represent regional financial service issues.

- **Consumer Tagging Accuracy:** A significant proportion of complaints are tagged with an “unknown” label, limiting the ability to analyze consumer demographics accurately.
- **Consumer Narrative Bias:** The dataset relies on consumer-submitted narratives, which may introduce subjectivity and bias in the overall sentiment of complaints since the majority of consumers do not submit one.
- **Company-Specific Focus:** Companies with larger market shares are likely to receive more complaints, which could unfairly highlight them as underperforming compared to smaller competitors.
- **External Factors:** The dataset does not provide context about external factors, such as economic downturns, policy changes, or public scandals, which could influence spikes in complaints.
- **Temporal Patterns:** The dataset starts in the middle of June and ends in the middle of September.
- **Public API:** Limited amount of shared user data for privacy concerns.
- **Digital Divide:** Consumers without access to the CFPB platform to submit complaints allows for potential demographic bias.

Building the Dashboard: Empowering Stakeholders

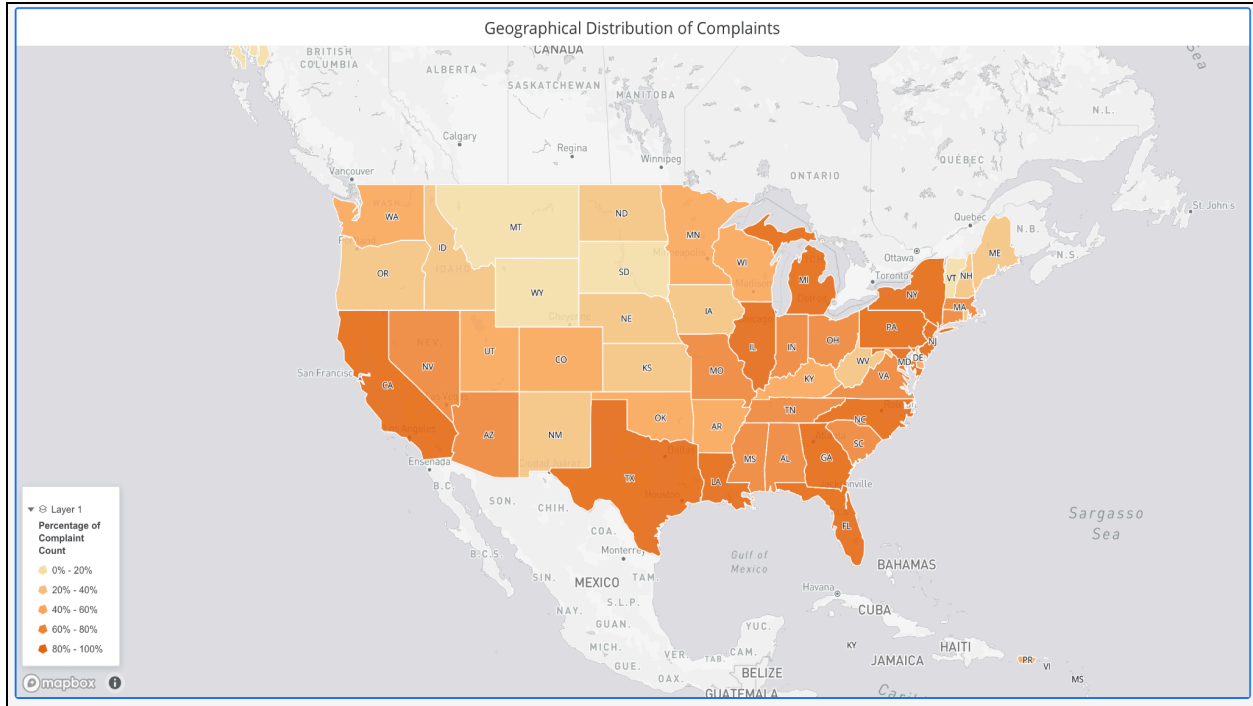
With MicroStrategy's powerful visualization platform and Python's data processing capabilities, we developed a dashboard that translates raw data into actionable insights. Here's how it works:

1. Main Page Overview



The homepage provides a high-level summary of the dataset:

- **Complaint Count:** Over 626,000 complaints filed across 1,900 companies within a 93-day period, averaging 6,737 complaints per day.
- **Geographic Trends:** An interactive map displays the percentage of complaints by state, helping stakeholders identify areas with higher complaint volumes.



- **Top Companies by Complaint Count:** A bar chart ranks companies, with TransUnion, Equifax, and Experian – major players in credit reporting – at the top, highlighting those receiving the highest complaint volumes.
- **List of Products:** Provides a searchable list of financial products involved in complaints, including credit cards, mortgages, and debt collection issues.

This page serves as a launch point for deeper analysis into complaint trends, empowering users to explore data by product, issue, and company.

2. Complaint Analysis



This section dives deeper into trends and patterns:

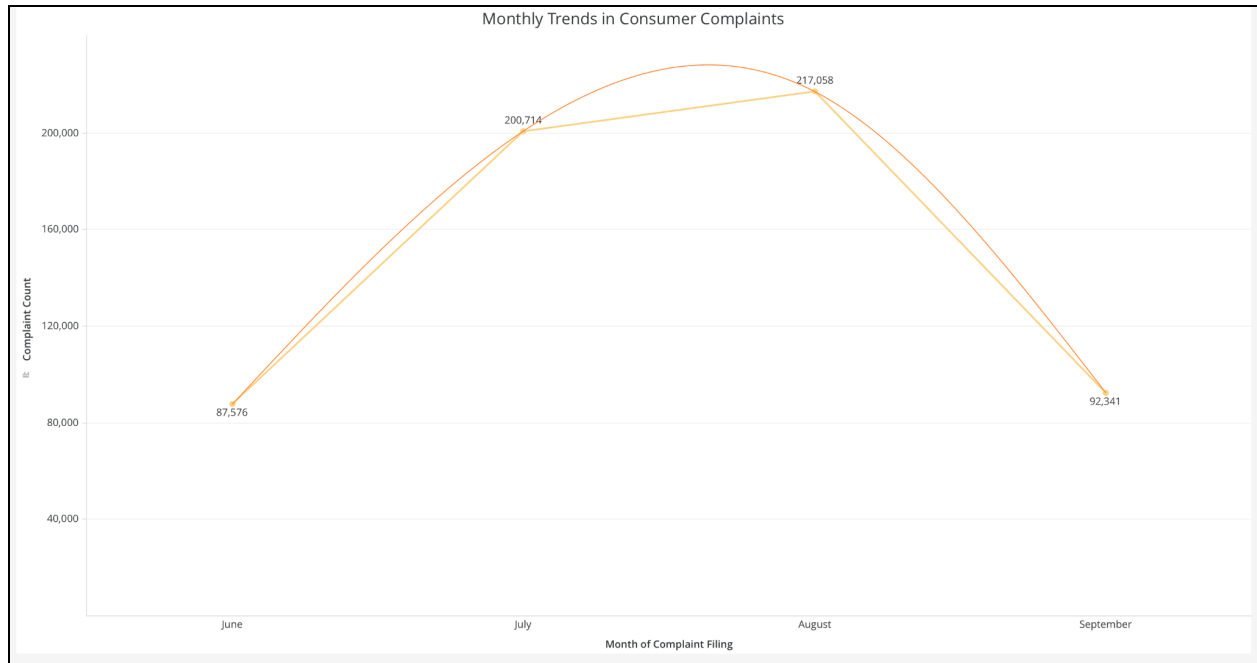
- **Complaint Counts by Issue:** Highlights the most common complaints, with "Attempts to collect debt not owed" leading with over 14,000 complaints, significantly outpacing other categories.
- **Geographic Trends:** An interactive map visualizes the distribution of complaints by state, enabling stakeholders to identify regions with the highest density of issues.
- **Top Products by Complaints:** A bar chart shows credit reporting and debt collection as the most problematic financial products, reflecting systemic challenges in these areas.
- **Top Companies with Timely Responses:** A chart ranks companies based on their timeliness in responding to complaints, providing benchmarks for customer service.
- **Company Responses to Complaints:** Displays the distribution of resolution methods, including monetary relief, explanations, and unresolved cases.

This analysis empowers stakeholders to target systemic issues, identify high-risk areas, and assess company performance in addressing consumer concerns.

3. Consumer Analysis



- **Complaint Distribution Amounts:** Visualizes complaints by product, highlighting checking or savings accounts at only 1.7%, compared to higher volumes in other categories.
- **Consumer Sentiments:** Breaks down sentiment distribution with 10.28% positive, 8.44% negative, 2.4% neutral, and 78.88% marked as unknown.
- **Consumer Narratives:** Provides real examples of complaints, revealing systemic issues such as "incorrect information" or unresolved disputes with credit agencies. These are further emphasized in the word cloud, where terms like "report," "improper use," and "company investigation" dominate.
- **Customer Complaint Tags:** Segments complaints by specific groups, such as servicemembers and older Americans, enabling targeted analysis of demographic-specific issues.
- **Monthly Trends in Consumer Complaints:** Tracks complaint volume over time, with a large increase in August of over 17,000 complaints. Important to note that the data has a start and end date in the middle of the months of June and September.



This analysis provides actionable insights into consumer experiences and key issues across financial products and demographics.

The Outcome: Actionable Insights

Our dashboard delivers actionable insights to empower stakeholders and address key challenges:

1. Highlighting High-Complaint Categories

- Credit reporting and debt collection stand out as the top issues, indicating systemic challenges in these areas.

2. Geographic Trends

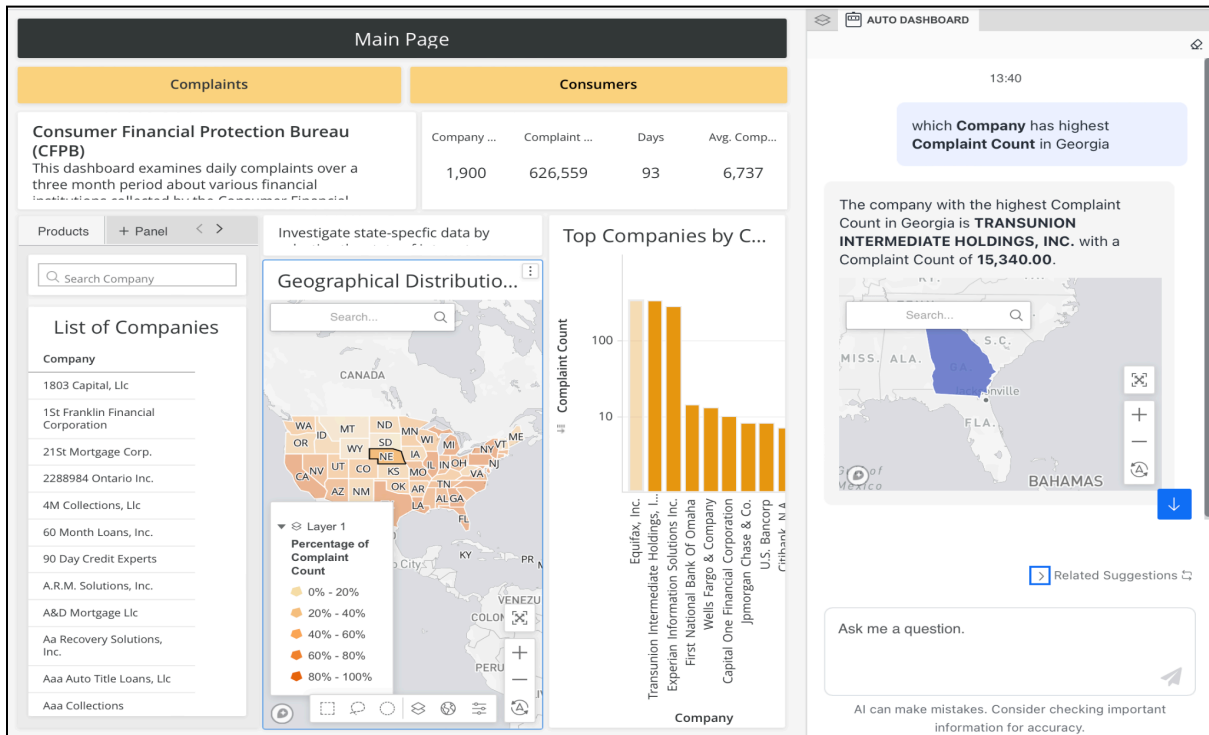
- Regional analysis identifies states like California and Texas as hotspots for complaints, guiding policymakers to address localized issues.

3. Company Response Patterns

- Analysis of company responses reveals variability in how complaints are resolved, from monetary relief to explanations, or unresolved cases.

4. AI-Powered Insights

- The dashboard includes an AI bot that enhances accessibility by answering critical questions, such as:
 - Which Company has the highest Complaint Count in Georgia?



- What is the distribution of complaints by product for the top issue?
- What is the distribution of company responses for the top issue?

This feature allows users to interact with the data intuitively, uncovering targeted insights with ease.

Why It Matters

The dashboard bridges the gap between raw data and actionable insights, benefiting three key groups:

- **For Consumers**
 - Compare companies and financial products based on complaint trends.
 - Identify recurring issues to make informed decisions about financial services.
- **For Policymakers**
 - Spot systemic issues in industries like credit reporting or debt collection.
 - Use data-driven insights to design effective regulations.
- **For Companies**
 - Benchmark performance relative to competitors.
 - Address common consumer complaints to build trust and improve services.

What CFPB Can Do

While our dashboard unlocks significant insights, there is room for improvement in the CFPB dataset itself:

1. **Enhance Sentiment Tagging:** A significant portion of complaints is marked as “unknown,” limiting sentiment analysis. Advanced tagging could provide more accurate emotional insights.
2. **Standardize Data Fields:** Uniform formatting of complaint types and company responses would improve usability.
3. **Offer Consumer-Friendly Tools:** Dashboards like ours could be integrated directly into the CFPB website, broadening access to insights.

Conclusion

At its core, this project reflects our commitment to transparency and consumer empowerment. By harnessing the power of MicroStrategy's platform and Python's analytical capabilities, we've turned raw data into an accessible tool for change. The dashboard enables consumers to hold companies accountable, equips policymakers to tackle systemic issues, and helps companies improve their services.

This collaboration between MicroStrategy and our team demonstrates the potential of data-driven insights to transform the financial services landscape, making it fairer, more transparent, and more consumer-friendly.

Contact Information

For further details or assistance, please contact:

Karim Tarik Haidar Alami - karimalami@gwmail.gwu.edu

Rameen Ridah - rameen.ridah@gwmail.gwu.edu

Kennedi Neri - keneri13@gwmail.gwu.edu

Appendices

CFBP's Consumer Complaint Database:

https://www.consumerfinance.gov/data-research/consumer-complaints/search/?chartType=line&dateInterval=Month&date_received_max=2024-09-14&date_received_min=2024-06-14&lens=Product&searchField=all&subLens=sub_product&tab=Trends